Seminar 4

Last week a **type** was built from constructors. This week a **claim** is: `Buyable 13` holds because there is a way to derive it, and `Buyable 7` fails because there is none.

Two moves, and the homework is made of them: **build** a derivation, and **take one apart**. Round 1 is `≤`, straight from the homework. Rounds 2–5 are a rule set you have not seen — 3¢ and 5¢ stamps.

Round 1 — `≤` is three lines of rules

```
#check @Nat.le.refl   -- ∀ {n}, n.le n
#check @Nat.le.step   -- ∀ {n m}, n.le m → n.le m.succ

theorem two_le_five_term : 2 ≤ 5 :=
  Nat.le.step (Nat.le.step (Nat.le.step Nat.le.refl))

theorem two_le_five : 2 ≤ 5 := by
  repeat constructor
```

the same derivation, twice

```
-- 2 ≤ 2   refl
-- 2 ≤ 3   step
-- 2 ≤ 4   step
-- 2 ≤ 5   step
```

`repeat constructor` found that term by trying `refl`, then `step`, at every goal. It is proof *search*, and slide 8 is where it stops working.

`zero_le'` — nothing to take apart

```
theorem zero_le' (n : Nat) : 0 ≤ n := by
  induction n with
  | zero => exact Nat.le.refl
  | succ n ih => exact Nat.le.step ih
```

the `succ` goal

```
n : ℕ
ih : 0 ≤ n
⊢ 0 ≤ n + 1
```

Induction on the number, because `0 ≤ n` is what we must build, not something we were handed. When the derivation is a hypothesis instead, induct on *it* — round 4.

`le_of_succ_le_succ` — homework ex03

```
theorem le_of_succ_le_succ (n m : Nat) :
    n + 1 ≤ m + 1 → n ≤ m := by
  induction m with
  | zero =>
    intro h
    cases h with
    | refl => exact Nat.le.refl
    | step h' => cases h'
  | succ m ih =>
    intro h
    cases h with
    | refl => exact Nat.le.refl
    | step h' => exact Nat.le.step (ih h')
```

```
case zero.step                  case succ.step
h' : (n + 1).le 0               h' : (n + 1).le (m + 1)
⊢ n ≤ 0    -- no rule for this  ⊢ n ≤ m + 1   -- ih h' : n ≤ m
```

One of the four cases vanishes: `cases` on an impossible hypothesis leaves zero goals.

`le_myMax` — homework ex07, the `if`

```
def myMax (n m : Nat) : Nat :=
  if n ≤ m then m else n

theorem le_myMax (n m : Nat) :
    n ≤ myMax n m ∧ m ≤ myMax n m := by
  unfold myMax
  split
  · exact ⟨‹n ≤ m›, Nat.le_refl m⟩
  · exact ⟨Nat.le_refl n, Nat.le_of_not_le ‹¬ n ≤ m›⟩
```

after `split`

```
-- goal 1, with h✝ : n ≤ m
⊢ n ≤ m ∧ m ≤ m
-- goal 2, with h✝ : ¬n ≤ m
⊢ n ≤ n ∧ m ≤ n
```

`split` is `cases` for an `if`: each branch keeps the condition that selected it, inaccessibly named — `‹n ≤ m›` asks for it by type. `by_cases h : n ≤ m <;> simp [h]` does the same thing with a name you chose.

Round 2 — a rule set you have not seen

```
inductive Buyable : Nat → Prop where
  | zero : Buyable 0
  | add3 {n : Nat} : Buyable n → Buyable (n + 3)
  | add5 {n : Nat} : Buyable n → Buyable (n + 5)
```

What postage you can make from 3¢ and 5¢ stamps. Three rules, the same shape as `MyEven` — one base, two steps.

Build `Buyable 3`, `Buyable 9`, then `Buyable 13`.

The easy two

```
theorem buyable_three : Buyable 3 := .add3 .zero

theorem buyable_nine : Buyable 9 :=
  .add3 (.add3 (.add3 .zero))

-- the lecture's search, unchanged
example : Buyable 9 := by
  repeat first | exact Buyable.zero
    | apply Buyable.add3 | apply Buyable.add5
```

what `apply` had to solve

```
⊢ Buyable 9
-- apply Buyable.add3 needs
--   ?n + 3 =?= 9
-- unification: ?n := 6
```

A numeral is not a constructor pattern, and it does not have to be: Lean solves `?n + 3 =?= 9` by arithmetic. This is the same move that let `cases` work on `MyEven 3`.

The same search on 13

```
example : Buyable 13 := by
  repeat first | exact Buyable.zero | apply Buyable.add3 | apply Buyable.add5

error: unsolved goals
⊢ Buyable 1
```

```
13 --add3--> 10 --add3--> 7 --add3--> 4 --add3--> 1   stuck
13 --add5-->  8 --add5--> 3 --add3--> 0   ✓ (never tried)
```

`repeat` commits: it takes the first rule that applies and never comes back. `add3` applies to 13, so the search walks straight into 1 and gives up there. `aesop` fails on this goal too.

`buyable_thirteen`, two ways

```
theorem buyable_thirteen : Buyable 13 :=
  .add5 (.add5 (.add3 .zero))

example : Buyable 13 := by
  solve_by_elim [Buyable.zero, Buyable.add3, Buyable.add5]
```

the term, read outwards

```
.zero         -- 0
.add3 ...     -- 3
.add5 ...     -- 8
.add5 ...     -- 13
```

Either you know the answer (13 = 3 + 5 + 5, so write it), or you ask for a search that backtracks. `solve_by_elim` retries the other rules; `repeat` never does.

Round 3 — take a derivation apart

`¬ Buyable 1`, `¬ Buyable 4`, `¬ Buyable 7`.

There is nothing to compute here yet and no formula to appeal to — only the question **which rule could have concluded this**. `cases` asks it, and keeps only the answers.

Draw the tree for 7 on paper first: two rules can conclude it.

One level, then two

```
theorem not_buyable_one : ¬ Buyable 1 := by
  intro h
  cases h

theorem not_buyable_four : ¬ Buyable 4 := by
  intro h
  cases h with
  | add3 h₁ => cases h₁
```

why `cases h` alone finishes 1

```
-- zero:  1 =?= 0        ✗
-- add3:  1 =?= ?n + 3   ✗
-- add5:  1 =?= ?n + 5   ✗
-- no rule survives ⇒ no goals
```

For 4 exactly one rule survives (`4 = 1 + 3`), and it hands back `Buyable 1` — the previous theorem's statement, one `cases` away from done.

`¬ Buyable 7` — the tree, and the real goals

```
theorem not_buyable_seven : ¬ Buyable 7 := by
  intro h
  cases h with
  | add3 h₄ =>
    cases h₄ with
    | add3 h₁ => cases h₁
  | add5 h₂ => cases h₂
```

```
case add3                  case add3.add3             case add5
h₄ : Buyable (0 + 4)       h₁ : Buyable (0 + 1)       h₂ : Buyable (0 + 2)
⊢ False                    ⊢ False                    ⊢ False
```

Two branches at the root (7 = 4 + 3 and 7 = 2 + 5), one below `add3`, none below the rest. Four `cases` and the whole search space is exhausted — that is what "7 is not buyable" means. Slide 22 gets the same fact for free.

Round 4 — induct over a derivation

```
buyable_add         : Buyable n → Buyable m → Buyable (n + m)
buyable_three_mul   : Buyable (3 * a)
buyable_five_mul    : Buyable (5 * b)

buyable_iff_exists : Buyable n ↔ ∃ a b, n = 3 * a + 5 * b
```

`cases` gave one goal per rule. `induction` gives the same goals **plus an `ih`** wherever a rule had a premise. The rules are the cases; that is all there is to it.

`buyable_add` — induct on `hm`, not on `m`

```
theorem buyable_add (n m : Nat)
    (hn : Buyable n) (hm : Buyable m) : Buyable (n + m) := by
  induction hm with
  | zero => simpa using hn
  | add3 _ ih =>
    rw [← Nat.add_assoc]
    exact Buyable.add3 ih
  | add5 _ ih =>
    rw [← Nat.add_assoc]
    exact Buyable.add5 ih
```

the `add3` case

```
hn : Buyable n
ih : Buyable (n + n✝)
⊢ Buyable (n + (n✝ + 3))
-- rw [← Nat.add_assoc]
⊢ Buyable (n + n✝ + 3)
```

The goal is one `add3` away from `ih` — but only once the brackets match. This is the lecture's `even_add` with two step rules instead of one.

`buyable_three_mul` — back to induction on a number

```
theorem buyable_three_mul (a : Nat) : Buyable (3 * a) := by
  induction a with
  | zero => exact Buyable.zero
  | succ a ih =>
    rw [Nat.mul_succ]
    exact Buyable.add3 ih
```

the `succ` goal

```
ih : Buyable (3 * a)
⊢ Buyable (3 * (a + 1))
-- rw [Nat.mul_succ]
⊢ Buyable (3 * a + 3)
```

No derivation was given, so there is none to induct on — `a` it is. Same for `buyable_five_mul`, with `add5`. The lecture's `even_two_mul`, one rule over.

`buyable_iff_exists` — rules against a formula

```
  constructor
  · intro h
    induction h with
    | zero => exact ⟨0, 0, rfl⟩
    | add3 _ ih =>
      obtain ⟨a, b, hab⟩ := ih; exact ⟨a + 1, b, by omega⟩
    | add5 _ ih =>
      obtain ⟨a, b, hab⟩ := ih; exact ⟨a, b + 1, by omega⟩
  · rintro ⟨a, b, rfl⟩
    exact buyable_add _ _ (buyable_three_mul a) (buyable_five_mul b)
```

Left to right: every rule that built the derivation adds one stamp to the count, so the induction writes the formula down as it goes. Right to left: no induction at all — three lemmas already did it, and `3 * a + 5 * b` is literally `buyable_add` applied to two of them.

Round 5 — when no rule steps by one

Every amount from 8¢ up is buyable. Try it directly:

```
theorem buyable_add_eight (n : Nat) : Buyable (n + 8) := by
  induction n with
  | zero => exact .add5 (.add3 .zero)
  | succ n ih => ?_
-- ih : Buyable (n + 8)
-- ⊢  Buyable (n + 1 + 8)
```

`ih` is one cent short and **no rule adds one cent**. The statement is too weak to carry itself forward.

Carry three at once

```
theorem buyable_eight_nine_ten (n : Nat) :
    Buyable (n + 8) ∧ Buyable (n + 9) ∧ Buyable (n + 10) := by
  induction n with
  | zero => exact ⟨.add5 (.add3 .zero),
                   .add3 (.add3 (.add3 .zero)),
                   .add5 (.add5 .zero)⟩
  | succ n ih =>
    obtain ⟨h8, h9, h10⟩ := ih
    exact ⟨h9, h10, Buyable.add3 h8⟩
```

the `succ` goal

```
h8 : Buyable (n + 8)   -- ... h9, h10
⊢ Buyable (n + 1 + 8)
  ∧ Buyable (n + 1 + 9)
  ∧ Buyable (n + 1 + 10)
```

Two thirds of the step is shifting what you already have; the new third is `add3` on `h8`, since `n + 11 = (n + 8) + 3`. A stronger statement is easier to prove — it hands the induction more to work with.

From `8 ≤ n` to a derivation

```
theorem buyable_of_le (n : Nat) (h : 8 ≤ n) : Buyable n := by
  obtain ⟨k, rfl⟩ := Nat.exists_eq_add_of_le h
  rw [Nat.add_comm]
  exact (buyable_eight_nine_ten k).1
```

after `obtain`

```
k : ℕ
⊢ Buyable (8 + k)
-- rw [Nat.add_comm]
⊢ Buyable (k + 8)
```

`Nat.exists_eq_add_of_le` is the lecture's `le_iff_exists_add_eq` under its library name: a `≤` derivation turned into an equation, so `rfl` can substitute it away.

`buyable_iff` — the complete answer

```
theorem buyable_iff (n : Nat) :
    Buyable n ↔ n = 0 ∨ n = 3 ∨ n = 5 ∨ n = 6 ∨ 8 ≤ n := by
  constructor
  · intro h
    obtain ⟨a, b, rfl⟩ := (buyable_iff_exists n).mp h
    rcases a with _ | _ | _ | a <;> rcases b with _ | _ | b <;> omega
  · rintro (rfl | rfl | rfl | rfl | h)
    · exact .zero
    · exact .add3 .zero
    · exact .add5 .zero
    · exact .add3 (.add3 .zero)
    · exact buyable_of_le n h
```

Left to right is the formula doing the work; right to left is five derivations, four of them written out by hand and the fifth from the previous slide.

Where `omega` stops

```
-- with a bare `omega` in place of the rcases line:
error: omega could not prove the goal:
a possible counterexample may satisfy the constraints
  d ≥ 0
  c ≥ 0
  1 ≤ 3*c + 5*d ≤ 2
where
 c := ↑a
 d := ↑b
```

`omega` decides linear arithmetic; "which numbers have the form 3a + 5b" is not that — it cannot see that 1¢ and 2¢ are out of reach. So split `a` into 0, 1, 2, ≥3 and `b` into 0, 1, ≥2 by hand: `rcases a with _ | _ | _ | a <;> rcases b with _ | _ | b <;> omega`. Every one of the twelve cases left is linear.

`Decidable` — the rules, compiled away

```
instance (n : Nat) : Decidable (Buyable n) :=
  decidable_of_iff _ (buyable_iff n).symm

example : Buyable 100 := by decide
example : ¬ Buyable 7 := by decide
#eval decide (Buyable 7)   -- false
```

what the instance is

```
-- buyable_iff n : Buyable n ↔ (a decidable claim)
-- decidable_of_iff transports the decision
-- procedure across the ↔
```

Round 3 spent four `cases` on `¬ Buyable 7`. With the characterisation proved, `decide` settles it — and 100, and anything else — by computing. Same bridge as the lecture's `Decidable (MyEven n)`.

What to take away

- **A rule set gives you both directions.** To prove the claim, build a derivation; to use one, `cases` it. Rules that cannot conclude the goal simply are not there.
- **`induction` on a derivation, not on the number** — whenever the derivation is what you were handed (`buyable_add`, `buyable_iff_exists`).
- `repeat constructor` is greedy, not a search: it walks 13 → 10 → 7 → 4 → 1 and stops. Write the term, or use something that backtracks.
- **Strengthen the statement when no rule steps by one.** `Buyable (n + 8)` alone cannot carry an induction; the triple can.
- Rules → formula → computation is the lecture's triangle again: proving `buyable_iff` is what buys `by decide`.